Specification Document

Landing Solutions

For Panther Cloud Air

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Introduction

The intended outcome of Panther Cloud Air's proposed project is to build, test, and research the validity behind their airline operations. This will be done by developing a simulation that models the logistics and financial challenges of running an airline network. The simulation should encompass generating flights, booking tickets, tracking hours, airplane maintenance, fuel, and more. The simulation looks to test the routing airplanes must take, the hubs required to arrive and depart, but most importantly getting above airplane and airport costs. By taking in a holistic view of the airline operations in a simulation, this allows Panther Cloud Air to visualize performance and probability before having to commit to a large scale influx of capital.

This project also applies software engineering concepts to a real world problem. We are combining scheduling logic, simulation modeling and database integration in a team environment while building functional prototypes that can be used in the planning and decision making of airline operations.

Design Consideration

Assumptions and Dependencies

The following assumptions outline the expected conditions necessary for the successful operation and performance of the Panther Cloud Air system;

- Internet Connectivity

All users, including employees and administrators, are assumed to have stable internet access. This is critical, as all interactions with the Panther Cloud Air interface and backend services occur via online connections.

- Account Provisioning

Each user will be assigned a valid login credential prior to system usage. This ensures secure and controlled access to authorized system features and data.

- Language Proficiency

Users are expected to read and write in English. System interactions, error messages, and user documentation will be presented primarily in English.

- Input Compatibility

The system is designed to interpret and respond effectively to logical English input. Users are expected to follow conventional input structures when interacting with the system.

- System Reliability

When executed within the designated environment, both the simulation and production system are expected to run without failure, assuming all configuration and runtime dependencies are met.

The primary system dependency is Docker, which provides a containerized environment for deploying and running the server. Docker enables consistent setups across different environments (development, testing, and production), ensuring isolation and predictable performance.

Each assumption has been validated through testing and prototyping phases. Network reliability, authentication flows, language parsing, and containerized deployment were all verified under controlled conditions. These confirmations allow the team to proceed confidently with the expectation that the system will perform as intended when deployed under the described conditions.

Goals and Guidelines

Landing Solutions aims to ensure that the Panther Cloud Air system reflects the organization's core values of efficiency, scalability, and data integrity. Our objective is to design and implement a functional simulation capable of accurately modeling airline operations within a controlled, iterative environment.

Our team is dedicated to;

- Develop a realistic simulation featuring 4 operational hubs and 56 aircraft, accurately representing the logistics and operational structure of Panther Cloud Air.
- Ensure the system architecture accommodates the maximum number of company employees who rely on it for management, scheduling, and performance tracking tasks.
- Put user accessibility as a central design priority. Implement a dynamic profit-tracking system designed to maximize efficiency and revenue insights.

The system will remain fully editable throughout the 14-day simulation window, allowing administrators to test and adjust performance metrics as needed.

The simulation will not collect or interfere with actual passenger, employee, or administrative personal data. Likewise, no simulated data will persist or be manipulated beyond the defined 14-day operational window. This constraint maintains both ethical and security standards during testing and demonstration phases.

Our team follows the KISS (Keep It Simple, Stupid) principle throughout development. By prioritizing clarity, simplicity, and functionality, we ensure that system components remain maintainable, user-friendly, and easy to adapt during future expansion or testing phases.

Development Methods

The development team adopted the Spiral Model as the guiding methodology for the project. This approach allowed for iterative refinement through successive phases of planning, design, implementation, and evaluation. By integrating risk analysis and user feedback at each stage, we ensured that every iteration produced measurable improvements toward a functional and reliable prototype.

Each cycle of the spiral model emphasizes the following;

- Identifying core system needs, including simulation accuracy, user accessibility, and database reliability.
- Creating UI mockups and architecture diagrams to refine usability and workflow logic.

- Developing a working prototype with a focus on database integrity, backend robustness, and front-end simplicity.
- Conducting internal testing of both the simulation and user interface to ensure performance meets the project's operational goals.

To support efficient teamwork and project management, **ClickUp** and **Discord** were used for constant planning, communication, and task documentation. **Canva** facilitated early design prototyping and visualization of UI layouts. **SourceTree** integrated seamlessly with **GitHub**, enabling effective version control and repository management.

All significant modifications, commits, and updates were tracked and made visible within the GitHub repository, ensuring team-wide transparency and traceability throughout development.

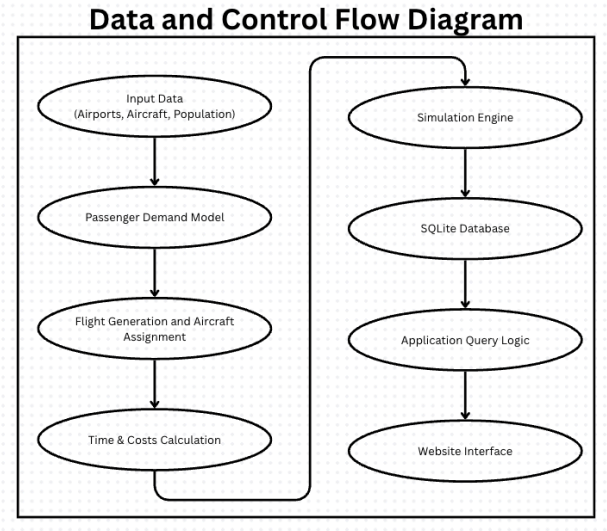
Architectural Design

Data and Control Flow

Our simulation begins with the input data that we're given (which is airport information, aircraft specification and metro population data). This data is then used by a passenger demand model to generate potential flight routes between airports based on expected travel demand. The equation for this demand model is:
Demand from City A to City B = (Population of A×0.005×0.02) × (Population B / Total population of all travelers)

Once the possible routes are generated, the system will assign aircraft to each flight based on availability and its operation constraints. Then the system calculates the flight distance, duration and any associated costs (like fuel consumption and airport fees). After all of the "scheduling phases", the simulation engine will apply any of the delays or other constraints based on the fixed delays or cancellations from the Project Proposal.

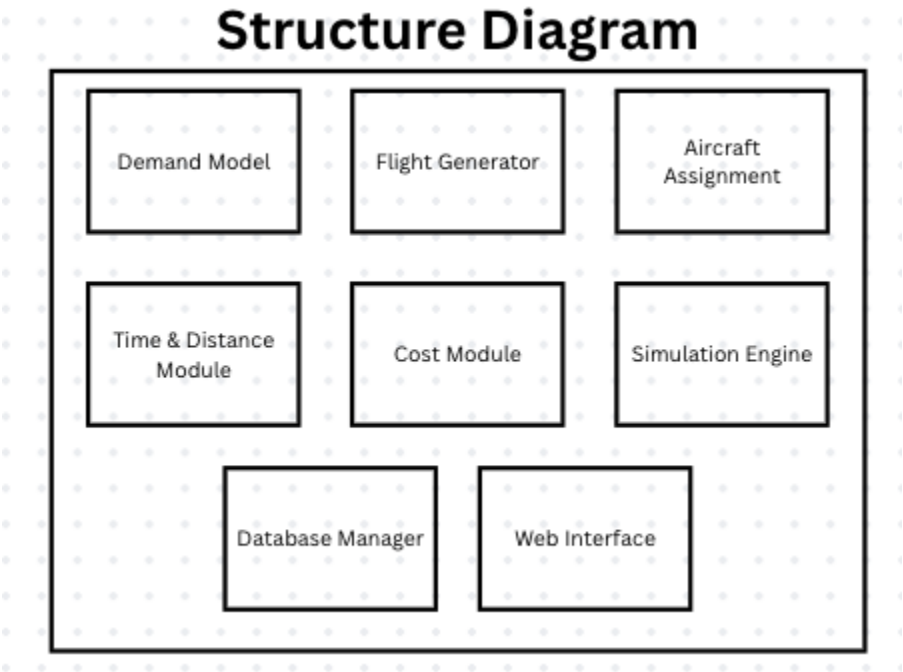
After all the scheduling and simulation days are populated, the corresponding flight records and passenger counts will be written to an SQL database. When a user interacts with the website (for example, searching for flights by a certain origin airport or simulation day), the request is processed by the application, which queries the database and retrieves flight records. The results are then returned to the user and displayed on the interface.



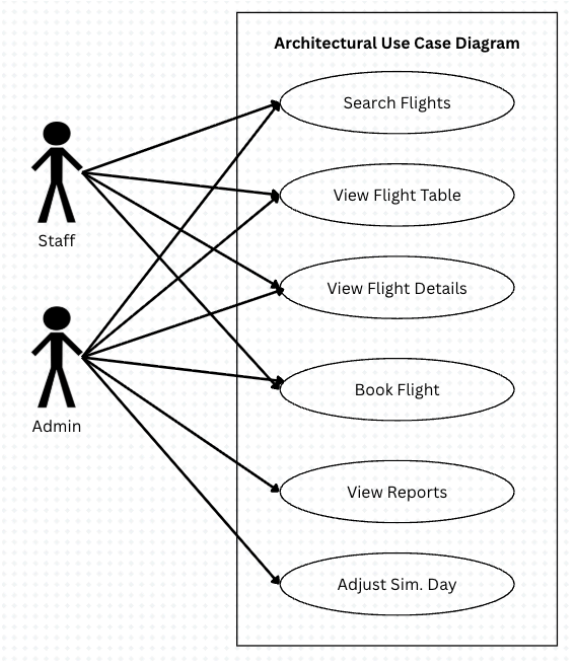
Derived Program Structure

The system has multiple functions that contribute to the airline simulation and data processing pipeline. Here are each of the components of the system as discussed earlier:

- Passenger Demand Model: Generates potential flight routes based on metro population proportions and travel demand assumptions.
  Demand from City A to City B = (Population of A×0.005×0.02) × (Population B / Total population of all travelers)
- Flight Generation Module: Creates flight instances using the output of the demand model, defining origin, destination, and initial scheduling parameters.
- Aircraft Assignment Module: Assigns aircraft to generated flights while considering aircraft availability and location constraints.
- Time and Distance Calculation Module: Computes flight durations using great-circle distance calculations and incorporates adjustments for operational factors such as wind effects.
- Cost Calculation Module: Determines operational costs for each flight, including fuel consumption and airport fees. These costs are used to calculate ticket pricing and overall profitability.
- Simulation Engine: Applies real-world operational conditions, including delays, cancellations, and maintenance requirements, across multiple simulation days.
- Database Management Module: Handles all interactions with the SQLite database, including inserting simulation results and retrieving data for user queries.
- Web Interface Module: Provides a user-facing interface that allows users to search for and view flight information stored in the database.



Use Cases


Architectural Use Case Diagram
Search Flights
View Flight Table
View Flight Details
Book Flight
View Reports
Adjust Sim. Day
Staff
Admin

Interface Design

Human Interface Specification

The important specifications for the human interface are, which user roles/personas do we have and the core actions these users are allowed to take. Since we changed our user personas it is important to acknowledge here that our group is no longer building support for a pilot role/persona after the client informed us that was not required. With that being said, we have 2 roles/personas, an employee role and an admin role. The employee role will be given to people like gate agents, they will be inside the airport and will use this website to provide any reasonable assistance to customers, like booking a flight, or giving general information about flights. The admin role will be given to a few select people, they will be able to visualize the simulation days, airplane maintenance statuses as well as overall financials.

Human Interface Design Rules

Landing Solutions main goal and rule set around human interface design is to follow **The Web Content Accessibility Guidelines (WCAG)** as this at its core is internationally reconsidered as these standards are built to ensure design is accessible to everyone, including people with disabilities. We take this a step further balancing white space and negative space, creating hierarchy with font sizing, and color to create a balanced and visually clear system taking user experience into account when building our user interface.

Rules:

- All buttons should have a default, hover, and active state
- All cards should have a reasonable padding to prevent text overlapping (0.25 - 1 rem | 4 to 16 px)
- All input fields should include a label and placeholder text to guide user input
- Any error message should be displayed clearly and use simple language
- Navigational elements should remain consistent in position, design, and hierarchy

External Interface Design

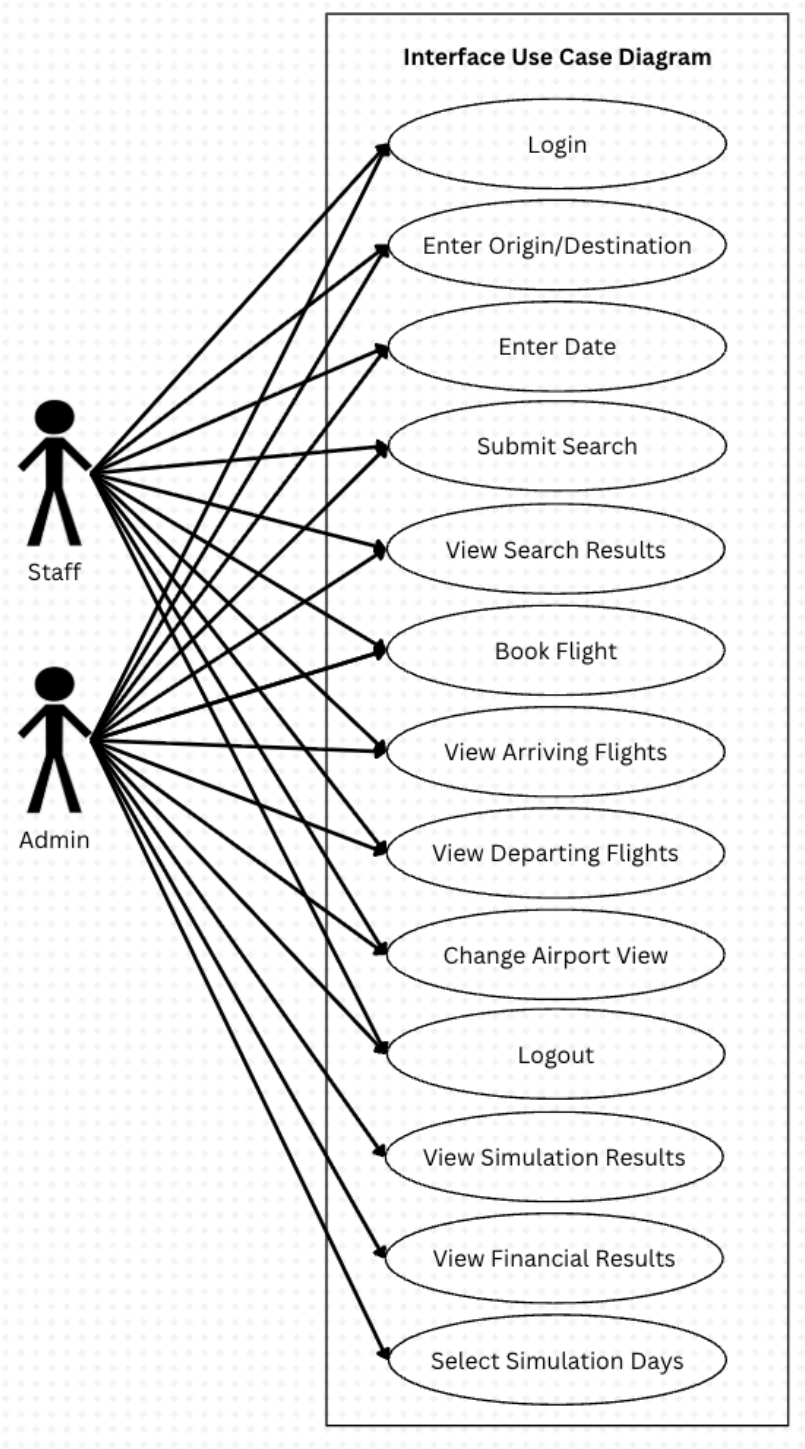
This will be an all encompassing external interface design scope.

External System: SQLite
Our database is going to be stored in SQLite, the purpose is to store and transfer user data in a structured way following a schema for repeat access.

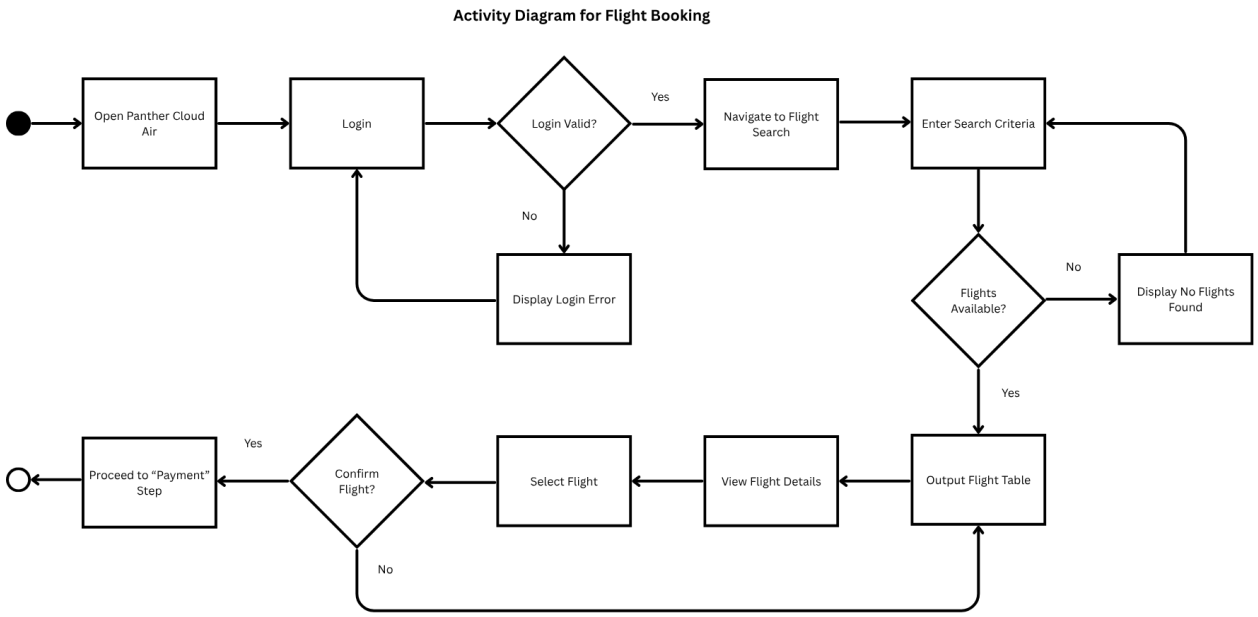
External System: SourceTree
SourceTree is a visual ui for git, it handles push, pull, and merges inside our github. This allows our group to work simultaneously and automatically handle version and state control and is designed to replace complex command-line instructions via a visual interface.

External System: Docker
Docker allows us to put our source code in a container, handling all the packages and dependency independently allowing a systematic and repeatable experience across all different hardware, and system differences.

Use Cases


Interface Use Case Diagram
Login
Enter Origin/Destination
Enter Date
Submit Search
View Search Results
Book Flight
View Arriving Flights
View Departing Flights
Change Airport View
Logout
View Simulation Results
View Financial Results
Select Simulation Days
Staff
Admin

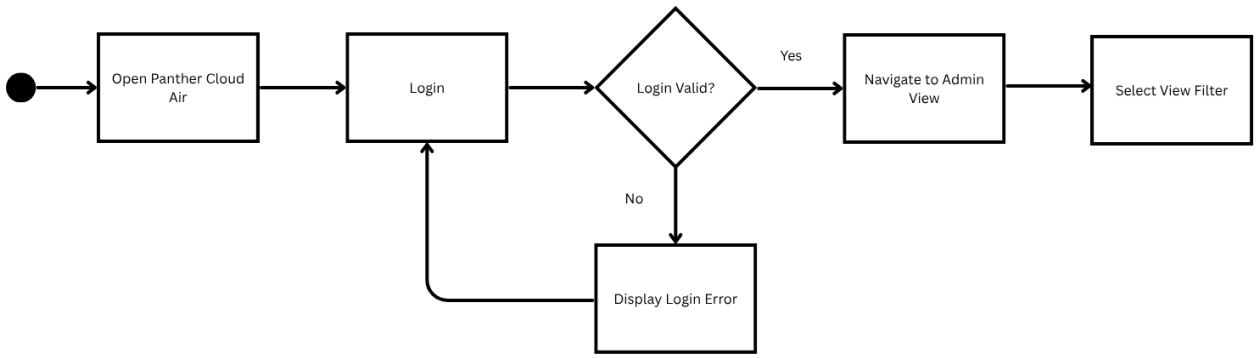
Procedural Design



Use Case Id	UC_01
Use Case Title	Book Flights
Actors:	Employee, Admin
Brief Description	Allows the employee and admin to book flights for passengers
Preconditions	Must have successful log-in
Flow	The employee or admin will first select the "Flight Selection" tab on the navigation bar. Next, using drop-down menus, they will select the required menus: "Departure Airport", "Arrival Airport", "Departure Date", "Arrival Date". Next, the employee or admin will be able to pick a specific flight from the table that will be filled based on the selected menus. Once here, they can confirm the flight they want to search and move onto the next page.
Exception Flow	If the flight search returns no flights due to invalid airports. If the employee/admin can not select a flight due to being full.

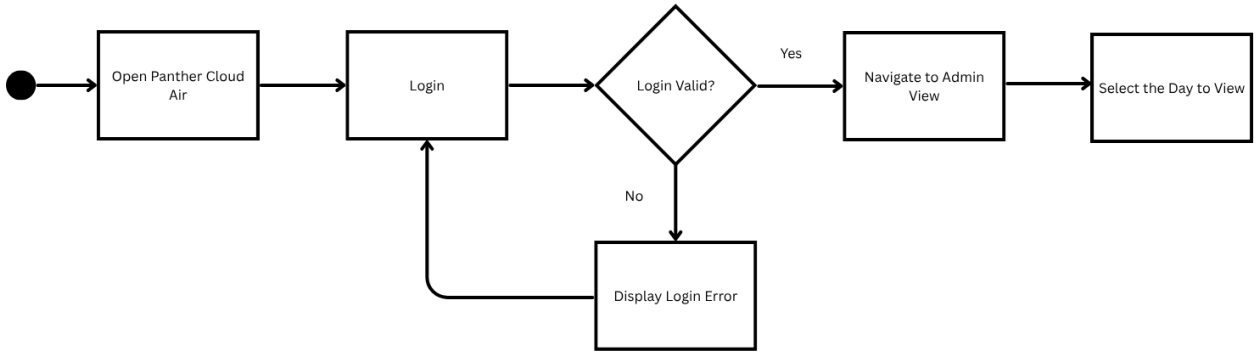
Use Case Id	UC_01
Postconditions	The selected flight information will be updated.

Activity Diagram for Revenue Reports

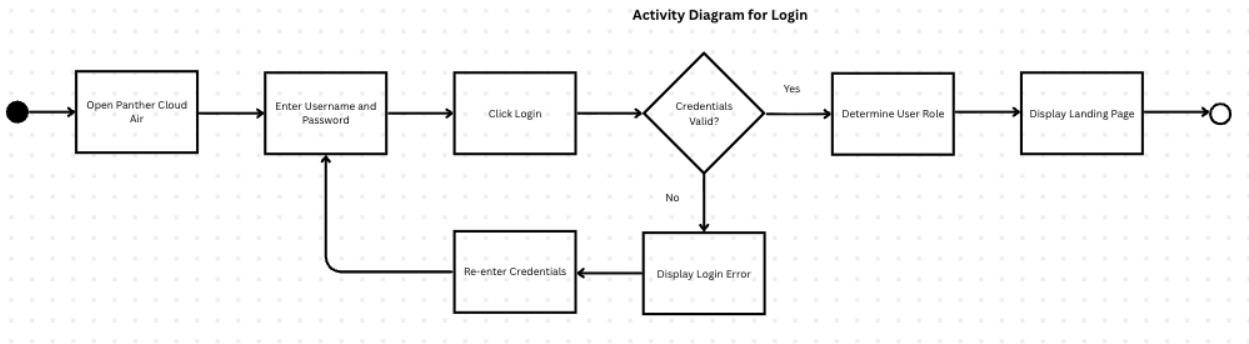


Use Case Id	UC_02
Use Case Title	View Reports
Actors:	admin
Brief Description	Allows the admin to view revenue reports
Preconditions	Must have successful log-in and be assigned the “admin” role
Flow	The admin will first select the “admin” view which will be located to the top right of the page. Once accessed, the admin will be able to view all revenue reports. The admin will further be able to filter the reports by pre-defined time amounts such as daily or monthly.
Exception Flow	If there is no report data available for the time period, the system will display an empty report message. If the database fails to laid the reports, the system will display a data retrieval error and prompt the admin to refresh or try again
Postconditions	The selected revenues reports are displayed on the screen based on the filters and can be adjusted dynamically

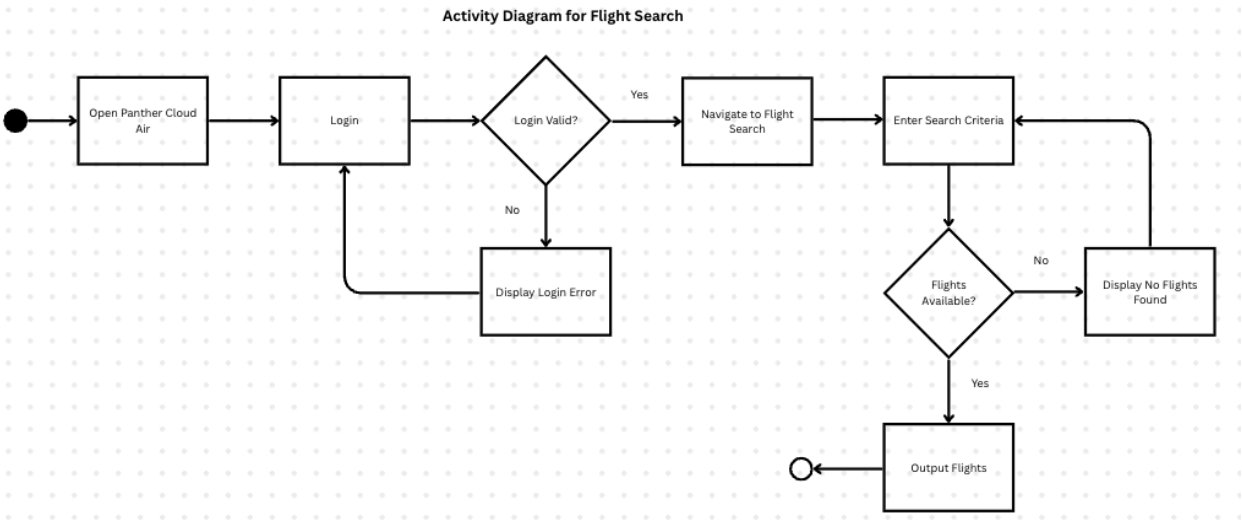
Activity Diagram for Adjusting Simulation Day



Use Case Id	UC_03
Use Case Title	Adjust Simulation Day
Actors:	admin
Brief Description	Allows the admin to change the day in the simulation.
Preconditions	Must have successful log-in and be assigned the “admin” role.
Flow	The admin will first select the “admin” view which will be located to the top right of the page. Once accessed, the admin will be able to view the “Day Selection” panel. Next the admin will be able to pick the specific day they want to view in the simulation.
Exception Flow	If the system fails to update the simulation due to a backend issue, the system will display an error and retain the previous simulation day
Postconditions	The simulation updates to the selected day and all relevant information is from the selected day

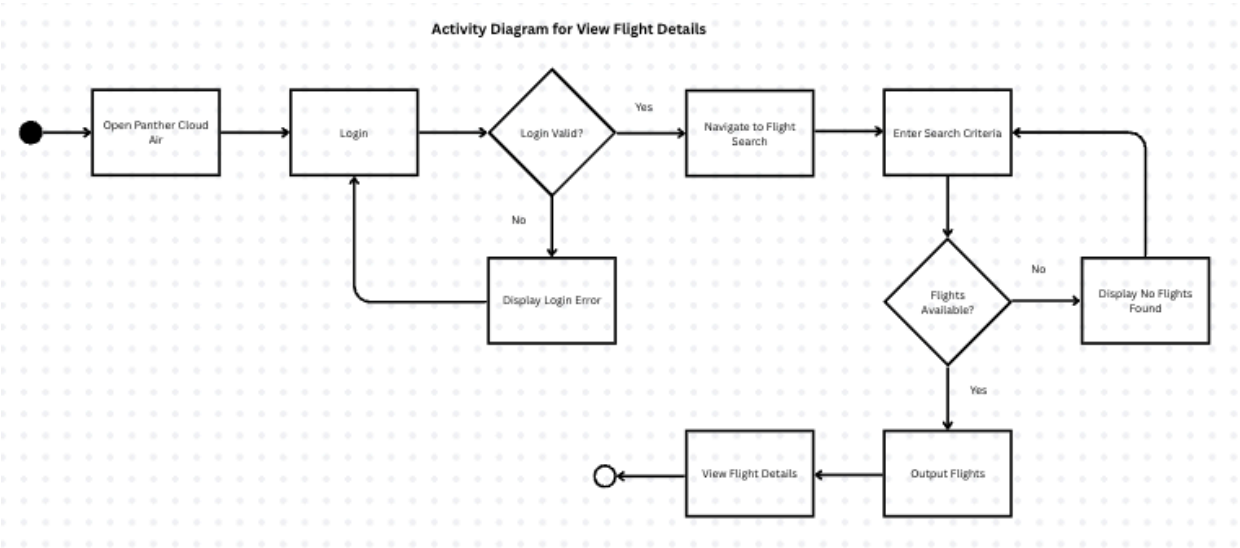


Use Case Id	UC_04
Use Case Title	Log in
Actors:	Staff, Admin
Brief Description	The user enters their username and password on the login page to access the system. Admin gets access to what the staff and the admin has access to (admin panel).
Preconditions	The system is running at localhost:3000. Then the user is able to log in with the right credentials to which the login page is shown
Flow	The user goes to localhost:3000. Then the system shows the login page with both username and password fields. The user enters both credentials, clicks login, then the system validates the credentials and the system redirects to the All Flights page.
Exception Flow	If the username or password is correct, then the system displays a login error message, the user remains on the login page, and the user can reenter credentials and try again. If the fields are left blank, the system prompts the users to fill in both fields
Postconditions	The user is successfully authenticated and redirected to our landing page which is our All Flights Page. Access permissions are assigned based on the users role and the nav bar and available features are updated accordingly.



Use Case Id	UC_05
Use Case Title	Search Flights
Actors:	Staff, Admin
Brief Description	The user can see all current PCA flights on the All Flights page. They can go though both the Arrivals and Departures to see incoming or outgoing flights along with the status.
Preconditions	The user is logged in to which the system has the flight data which is from the database
Flow	The user selects All Flights in the Navigation bar. The system loads the All Flights page which goes to the departures tab by default and queries the schedule table and displays flights in a table. Each row shows Origin, Time, Flight Number, Gate, and Status. The last of which is color coded. The user clicks the arrivals button to switch views. They can also click Departures to switch back. The system

Use Case Id	UC_05
	filters the table to show only the arriving flights
Exception Flow	No flights scheduled for a certain period, the system shows an empty table. If the database connection fails, then the system displays a connection error.
Postconditions	The All Flights table is displayed with current flight data. The active view is highlighted in blue. The user can click any flight number to navigate Flight Info.



Use Case Id	UC_06
Use Case Title	View Flight details
Actors:	Staff, Admin
Brief Description	The user can search up a flight by the flight number on the Flight info page. The system is also able to show the full breakdown which will include the departure info and inflight data.
Preconditions	The user is logged in and knows a valid flight number. The flight exists in the schedule table.
Flow	The user clicks Flight Info in the navigation bar, prompting the system to display flight information along with a search box. The user then types a flight number

Use Case Id	UC_06
	into the search field and presses Enter. In response, the system searches the database for the flight record and performs the necessary flight and cost calculations. Once processed, the system provisionally displays the flight details organized into four sections: Departure (including gate and scheduled time), In-Flight (covering takeoff and cruise speed), Arrival (with landing time and taxi time), and a Cost Summary (listing airport fees).
Exception Flow	If the flight number searched doesn’t exist, then the system shows a message that prompts the user to enter a different flight number. If the Flight number format is wrong, then the system shows them that the flight number is in the wrong format.
Postconditions	The flight detail card is shown with all four sections. Profit is shown in green at the bottom of the cost summary. The flight header shows the routes and status. The user can now search for another flight without leaving.

Updated MS Project Reports - Schedule and Costs

Schedule

Things are on track overall, even with a few adjustments in the schedule. The project started with planning and gathering requirements to understand what the client needed and what the system should do. After that, the focus shifted to building the interface and developing the main parts of the prototype. Work during the middle of the project included creating key pages like the login, schedule view, booking process, employee tools, and admin features. This is where there was more development on the frontend. Now, most of the main parts are built but needing to find the right cords to connect each part. This is important so that we don't fit square pegs into round holes. The system can show the basic workflow the client expects. The remaining work includes testing, fixing issues, finishing documentation, and getting ready for the final presentation which includes the fourth walkthrough.

Costs

The total cost of the project is mostly based on the team's time. Since this is a prototype, there aren't many extra expenses, so the main cost comes from development work. The team has 5 developers working about 7 hours per week over 12 weeks. That adds up to around 420 total hours. At $30 per hour, the total labor cost comes out to about $12,600. The work itself is split into phases like planning, design, building, testing, and documentation which are the core components of software engineering. So far, around 20 hours have been spent on requirements planning, which included figuring out what the client needs and what the system should do. There are also some equipment costs. The team used a Dell Pro 14 Premium PA 14250 laptop with an Intel Core Ultra 7 processor, 32 GB of RAM, and Windows 11. The total cost for the equipment is $1,659. Adding everything together, the total project cost is about $14,259.

Project Meeting Notes

To ensure transparency and alignment with client expectations, the following notes summarize key decisions and feature implementations derived from meetings conducted throughout the development phases. These items reflect confirmed client specifications and finalized adjustments integrated into the Panther Cloud Air system.

No additional security layers beyond hashed password storage within the system's database were required. This approach matches the client's specifications for internal data protection during simulation testing.

Landing Solutions operates as an independent development team without a dedicated IT support entity. This includes maintenance, deployment, and testing remaining internal to the development group.

The Pilot user role was removed from earlier prototypes following client approval. The finalized version of the system retains only two active user types: Admin and Employee, optimizing access control and interface consistency.

Enhancements were implemented in the Admin UI to improve navigation across different days within the simulation period. This update allows administrators to effectively review, track, and modify operations within the 14-day simulation window.

Our github repository will display summaries of each meeting we have had with the client if further clarification is needed.

Work Break Down

Finley Myers

Procedural Design - Use Case Tables for “Book Flight”, “View Reports”, “Adjust Sim Day”
Procedural Design - Activity Diagrams for “Book Flight”, “View Reports”, “Adjust Sim Day”
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Jack Jaramillo

Architectural Design
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Design Considerations
Project Meeting Minutes